TRADE THAT TIES: TRADE COMMUNITIES OVER TWO CENTURIES (1833–2025)

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Do not forget to introduce co-authors

Economic Historian

Post-doc in economic history

Now entrepreneur, formerly technical director of Médialab at SciencesPo. Our data scientist

INTRODUCTION

Caveat emptor

What do we call trade communities?

Why are we interested?

This is a work in progress. We are still in the garden of forking paths.

I would not mind impressing you, but more realistically I am here to share and learn.

Trade blocs, trade regions… groups of countries that trade more with one another than with the rest of the world according to some metric

Trade communities play a role in world trade

Gopinath et al. (2025), Bonadio et al. (2026), Bhagwati (1991…), Anderson and Norheim (1993),

Current rising international tensions have increased the interest for securing supply chains through trade fragmentation by geopolitical lines. Studies show the rise of friend-shoring, near-shoring and decoupling by trading more with geopolitical friends, as in these two recent Journal of International Economics papers

This is probably bad for global trade.

Bhagwati asked whether regional trade blocs were “building blocs or stumbling blocs”

It does not need to be so. Anderson and Norheim in an old paper (1993) even show that generally, regionalisation and globalisation are positively correlated.

And the story of "hyperglobalisation" from the 1980s to the 2000s, around the rise of the European Union, NAFTA, and Factory Asia can lead to a positive view of the formation of trade blocs.

Still, in the 1930s, with decoupling between trade blocs organised around formal and informal empires, gives less reason for optimism.

We look at this phenomenon with historical depth, which allows to show if the current period is unusual.

We look for trade communities with network science

Fagiolo et al. (2010) De Benedictis & Tajoli (2011), Piccardi & Tajoli (2012), Iapadre & Tajoli (2014)

Blocs might arise from reasons that are not linked to geopolitical tensions, and the contrast between different periods is interesting.

It is not only that countries trade with friends. It is also that you want to become friends with the countries you trade with. So trade communities arising from the bottom up, ie trade relations, are interesting to study.

In the early 1990s, it was not obvious with whom Eastern Europe would trade more. You could argue that trade created the community rather than the reverse.

We do not pre-impose trade blocs using network science. There is a small literature on that subject, but it has not really become mainstream.

We want to "break into the mainstream" by adding another step

Trade communities have economic consequences.

Lesger (2006), Greif (1993), Miller (2012), Rauch (1999, 2001), Rauch & Trindade (2002), Combes, Lafourcade & Mayer (2005), Chaney (2014)

T

The creation of trade communities is important because it has consequences for trade, in that it might encourage more trade.

But then you have an issue, because of course that relationship is deeply endogenous to the creation of communities.

Except for the extensive margin.

Look at the story of Amsterdam in the 16th and 17th centuries (Lesger 2006). From an actual entrepôt for goods to an information entrepôt that encouraged the formation of direct links between the Baltic and Western Europe for naval stores, grain, and various raw materials.

We argue this happened because of the creation of trade communities of people.

Ethnic work, eg. Greif's work on the Maghribi traders of the 10th-11th century

Double entendre in

Miller on the 20th European expansion created intercontinental networks of business and administrative people with shared overseas experience that encouraged trade

Shared business and financial networks give access to shared information through intermediaries.

What are the composition, determinants and effects of trade communities over two centuries ?

Outline

1. Data
2. Identifying trade communities
3. Finding the determinants of trade communities
4. Finding the effect of trade communities

I. DATA

DATA (1)

Well known IMF data from 1948

A database of bilateral world trade 1833-1938 (Girard et al. (2016))

RICardo project since 2004. 690,000 trade flows

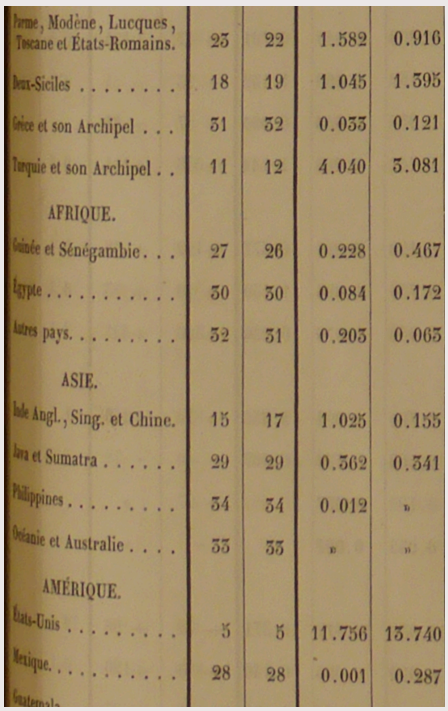
https://ricardo.medialab.sciences-po.fr/

All started with the discovery of the "Annales du Commerce Extérieur", an annual French source from the 1830s that reported bilateral trade statistics from numerous foreign countries

Girard, Paul, et al. « **RICardo Project : Exploring XIX Century International Trade** ». *Digital Humanities 2016: Conference Abstracts* [Agiellonian University & Pedagogical University, Kraków, Poland], 2016, p. 208-10, http://dh2016.adho.org/abstracts/177.


RICardo
Trade
Data
Glossary
About
FR | EN
The RICardo Project
Trade between Nations from c. 1800 to 1938
RICardo (Research on International Commerce) is a project dedicated to trade between nations over a period spanning the beginning of the Industrial Revolution to the eve of the Second World War.
It combines a historical trade database covering all of the world's countries and a website which invites to an exploration of the history of international trade through data visualizations.
IMPORTATIONS.
EXPORTATIONS.
161,067,000
97,639,000
72,467,000
38,245,000
24,007,000
20,867,000
14,813,000
8,860,000
8,591,000
7,182,000
5,043,000
105,305,000
123,504,000
84,937,000
14,635,000
5,207,000
10,248,000
8,050,000
6,352,000
Reporting
Reporting depicts the distribution of trade by partners of a reporting
Partner
Partner depicts an entity's trade as it is reported by the reportings
Bilateral
Bilateral focuses on trade between two entities and allows for a comparison between their trade statistics
World
World provides different historical series of world trade and its breakdown by entity

DATA (2)



Parme, Modène, Lucques, Toscane et États-Romains.	23	22	1.582	0.916
Deux-Siciles	18	19	1.045	1.395
Grèce et son Archipel . . .	31	32	0.033	0.121
Turquie et son Archipel . .	11	12	4.040	3.081
AFRIQUE.				
Guinée et Sénégambie . . .	27	26	0.228	0.467
Égypte	30	30	0.084	0.172
Autres pays.	32	31	0.203	0.063
ASIE.				
Inde Angl., Sing. et Chine.	15	17	1.025	0.155
Java et Sumatra	29	29	0.362	0.341
Philippines	34	34	0.012	»
Océanie et Australie	33	33	»	»
AMÉRIQUE.				
États-Unis	5	5	11.756	13.740
Mexique.	28	28	0.001	0.287

Issue with the sources: Heterogeneity of trade entities in trade archives

Tableau général du commerce avec les pays étrangers (Belgique, 1850)\

Cuba et Porto-Rico
Possessions anglaises

DATA (3) : TRYING TO BE FAITHFUL TO THE SOURCES

Source: RICardo database

Flow (1000 £)	year	expimp	reporting	partner
1427004.4	1850	Exp	Belgium	United Kingdom
1561.9	1850	Exp	Belgium	Africa (other)
55802.4	1850	Exp	Belgium	Austria-Hungary (Austrian Empire)
103170.1	1850	Exp	Belgium	Brazil
80560.1	1850	Exp	Belgium	Chile & Peru
55841.5	1850	Exp	Belgium	Cuba & Puerto Rico
11402.6	1850	Exp	Belgium	Denmark
42174	1850	Exp	Belgium	Two Sicilies
5584.1	1850	Exp	Belgium	Egypt Arab Republic (Egypt)
64002.9	1850	Exp	Belgium	Spain
377691.6	1850	Exp	Belgium	United States of America
2608227.6	1850	Exp	Belgium	France
62245.7	1850	Exp	Belgium	Luxemburg
3983.1	1850	Exp	Belgium	Greece & Ionian Is.
8512.8	1850	Exp	Belgium	Guatemala
31981.9	1850	Exp	Belgium	Gambia & Guinea (French Guinea) (Rivieres du Sud) & Senegal
2967.7	1850	Exp	Belgium	Haiti & Venezuela
220944.8	1850	Exp	Belgium	Hanover & Oldenburg
11207.3	1850	Exp	Belgium	India & People's Republic of China (China) & Singapore
7302.3	1850	Exp	Belgium	Java & Sumatra
6404.2	1850	Exp	Belgium	Mecklenburg Schwerin
26475.9	1850	Exp	Belgium	Mexico
6365.1	1850	Exp	Belgium	Australia & Oceania
31513.3	1850	Exp	Belgium	Lucca & Modena & Papal States & Parma & Tuscany
1221796.4	1850	Exp	Belgium	Netherlands
117.1	1850	Exp	Belgium	Philippines
2303.9	1850	Exp	Belgium	Portugal
0	1850	Exp	Belgium	British America
1261080.6	1850	Exp	Belgium	Prussia
13862.7	1850	Exp	Belgium	Argentina (La Plata)
95984.8	1850	Exp	Belgium	Russia (USSR)
151474.9	1850	Exp	Belgium	Piedmont & Sardinia
1757.2	1850	Exp	Belgium	Bavaria & Frankfurt & Saxony
29248.4	1850	Exp	Belgium	Norway & Sweden
30146.5	1850	Exp	Belgium	Switzerland
8201788.6	1850	Exp	Belgium	World as reported
126639.1	1850	Exp	Belgium	Archipelago & Turkey (Ottoman Empire)
14448.4	1850	Exp	Belgium	Hanse Towns

Source: TRADEHIST (Fouquin & Hugit 2016)

ISO_o	ISO_d	year	Flow Exp (1000 £)
BEL	2SICIL	1850	42174
BEL	ARG	1850	13862.8
BEL	AUTHUN	1850	55802.4
BEL	BRA	1850	103170.1
BEL	CHE	1850	30146.6
BEL	CHL	1850	5583.7
BEL	CUB	1850	55841.5
BEL	DNK	1850	11402.6
BEL	EGY	1850	5584.1
BEL	ESP	1850	7891.5
BEL	FRA	1850	4113120
BEL	GBR	1850	1520194
BEL	GRC	1850	3983.1
BEL	GTM	1850	8512.9
BEL	IDN	1850	7302.4
BEL	LUX	1850	62245.7
BEL	MAR	1850	2755.2
BEL	MECKL	1850	6404.2
BEL	MEX	1850	26475.9
BEL	NLD	1850	1929164.3
BEL	OTTO	1850	126639.2
BEL	PHL	1850	117.2
BEL	PRT	1850	2303.9
BEL	PRUS	1850	1261081
BEL	SEN	1850	31981.9
BEL	SWENOR	1850	29248.4
BEL	USA	1850	377691.6
BEL	USSR	1850	95984.9
BEL	ZOLL	1850	1325083.8

Les Îles Ioniennes sont un protectorat britannique jusqu'aux années 1860

DATA (4) STANDARDIZATION OF NAMES (INCLUDING TRANSLATION)

From c. 10,000 names to c. 2.500 names (COW : 1,200)

COW code	COW name	ISO code	ISO name
7501	Andaman and Nicobar Is.	-	-
242	Anhalt Cothen	-	-
5813	Anjouan	-	-
8152	Annam	-	-
7588	Baluchistan (Kalat)	-	-
80	Belize (British Honduras)	BLZ	Belize
434	Benin (Dahomey)	BEN	Benin
42	Dominican Republic (Santo Domingo)	DOM	Dominican Republic
404	Guinea-Bissau (Portuguese Guinea)	GNB	Guinea-Bissau

DATA (5)

There are very trading actors in the sources

RIC type	All entities	Reporting	Partner	Examples
Polity	530 (22.7%)	286	483	Bulgaria, Gambia, Singapore, Queensland…
Locality	707 (30.2%)	120	571	Asian Russia, Bahia, France (Atlantic Coast)…
Group	860 (36.8%)	37	797	Altona & Holstein, Trinidad & United Kingdom…
Colonial area	130 (5.6%)	0	120	British Africa, Portuguese Asia, Spanish West Indies…
Geographical area	112 (4.8%)	4	94	Central America, Eastern Africa, Persian Gulf…
Total	**2339**	**447**	**2065**	

To identify "British Africa" we need to know what are the British African colonies in a given year.

DATA (6)

A database of geopolitical entities of the world since 1816 (Dedinger & Girard (2021)

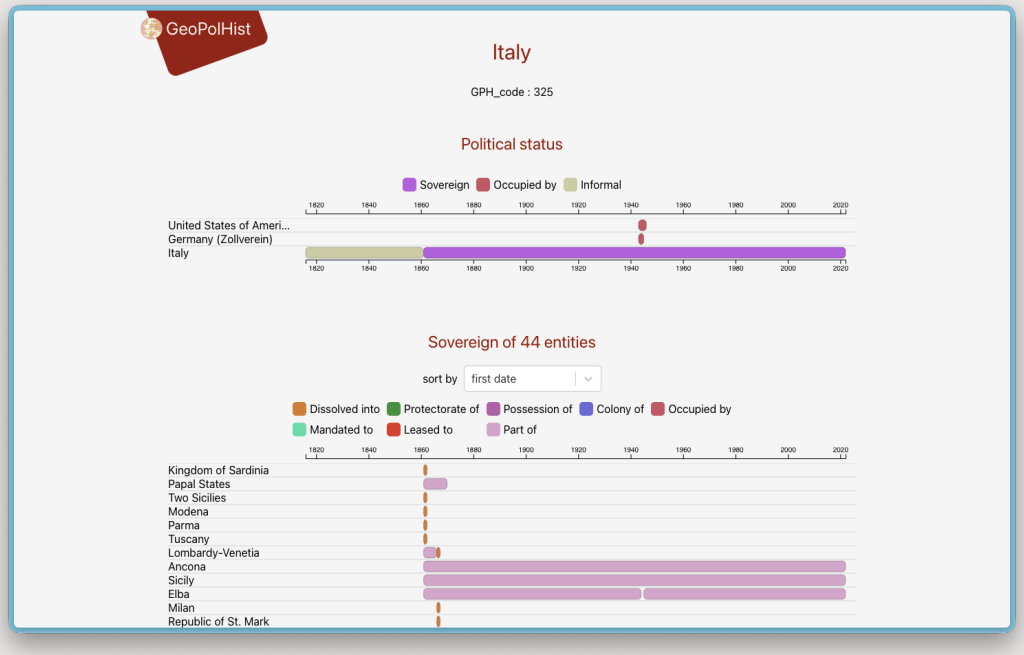
GeoPolHist

https://medialab.github.io/GeoPolHist/#/GeoPolHist/

DATA (7)
The GeoPolHist dataset

GPH code	GPH name	Start year	End year	GPH Status	GPH sovereign code
325	Italy	1816	1861	Informal	
325	Italy	1861	2022	Sovereign	
325	Italy	1943	1946	Occupied by	2
325	Italy	1943	1945	Occupied by	255
326	Trieste	1816	1918	Part of	300
326	Trieste	1918	1943	Part of	325
326	Trieste	1943	1947	Occupied by	2
326	Trieste	1947	1954	Mandated to	1
326	Trieste	1954	2022	Part of	325
327	Papal States	1816	1861	Sovereign	
327	Papal States	1861	1870	Part of	325
327	Papal States	1870	1870	Dissolved into	328
328	Vatican City	1816	1870	Part of	327
328	Vatican City	1870	1929	Part of	325
328	Vatican City	1929	1964	Sovereign (unrecognized)	
328	Vatican City	1964	2022	Sovereign	
329	Two Sicilies	1816	1860	Sovereign	
329	Two Sicilies	1860	1861	Part of	324
329	Two Sicilies	1861	1861	Dissolved into	325

GeoPolHist: Italy (325)



For each entity, we list their political status along with time and their links to sovereign parent entities.

DATA (8)

To build coherent bilateral trade matrices, one needs to reduce trading actors heterogeneity thanks to GeoPolHist data

1. identify non-polities, groups… informal entities.
2. aggregate non-polities entities (localities, part-of…) to their polities.
3. split trade of groups, geographical and colonial areas

DATA (9)

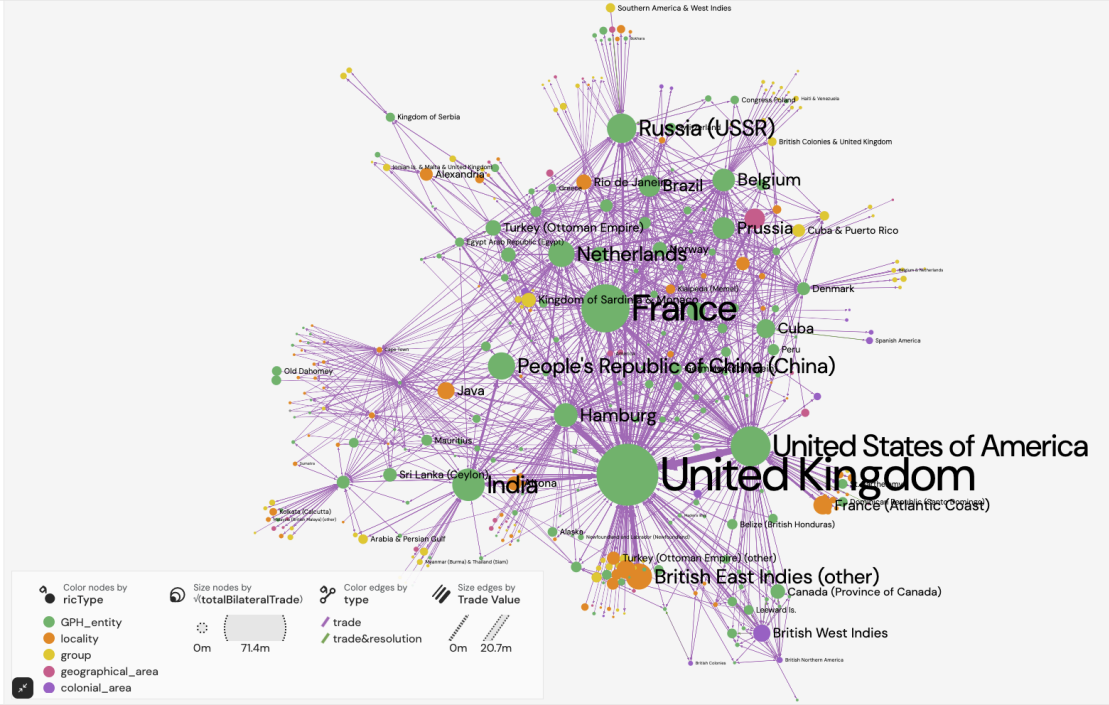
Method: Multilayer networks combining trade and politics

trade flow edges from **RICardo** dataset

geopolitical resolutions edges mainly from **GeoPolHist** dataset (« localities » , eg Bilbao, come from **RICardo**)

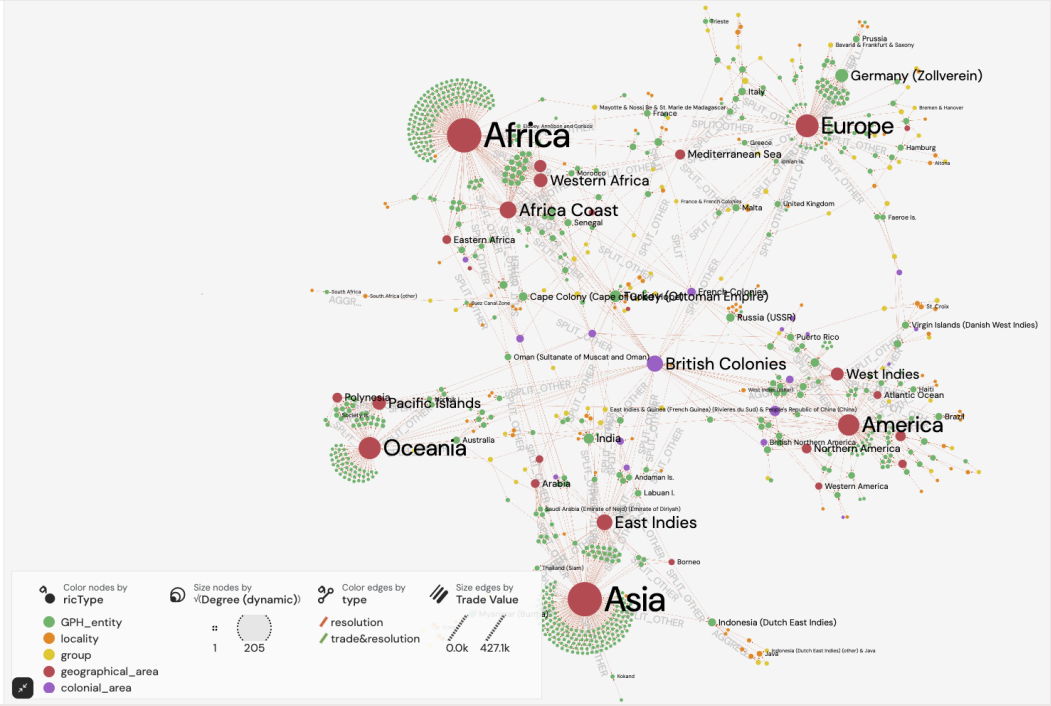
Our proposal is to merge those two dataset RICardo for trade, GeoPolHist into one multilayer network.

The 1850 trade network



This is the original 1850 trade network

The 1850 resolution network



This is the network of the resolution edges

Zoom to malta

mention Gephi lite

DATA (10)

Harmonization process

1. Polities resolution
2. Trade reporters aggregation/split
3. Trade partners aggregation
4. Trade partners splits with year ratio method
5. Trade reporters & partners splits with gravity model method

![===/assets/trade data model.png)

Bilateral trade data model

Note that we have trade between reporters but also between reporter and partners, i.e. entity mentioned in sources but for which we don't have a "own" source. They can be approached through mirror flows

TRADE PARTNER SPLITS WITH YEAR RATIO METHOD

Difficult task: how to decide the ratios to split one trade value into many?

We look into **adjacent years** (+/- 10-years window) networks\

TRADE PARTNER/REPORTER SPLITS WITH GRAVITY MODEL METHOD

We try to impute flows we couldn't split with the adjacent years method by using a ppml gravity model (Anderson et Van Wincoop (2003), Santos Silva & Tenreyro (2006)).

We do this cross-section. Very far from a balanced panel

for *dissociated* trade flows with the **same set of partners** for the same reporter.

If we find one compatible year, we calculate **split ratios** for that year and reapply those **on the original trade value**.

This process support partial split.
If a set of partners from a group is found as one + another group, the one found will be split, the rest will stay as a group.

Special cases: Areas

Areas (geographical or colonial) are implicit groups.

The composition of the group is not explicit in the source.

We use geographical or colonial sets which we adapt to the source context:

we remove all theoretical members of the area which are already cited by the reporter

we remove those that would not be part of the trade network otherwise (not directly cited by another source)

We use fixed effects on importers and exporters, geographical distance, geopolitical link existence (GeoPolHist) variables (to be extended).

The inferred values are used to compute a ratio which is then applied on the original values.

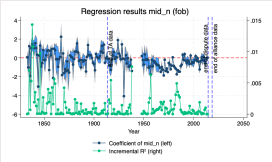
Anderson, James E., et Eric Van Wincoop. « Gravity with Gravitas: A Solution to the Border Puzzle ». *American Economic Review*, vol. 93, no 1, February 2003, p. 170-92. DOI.org (Crossref), https://doi.org/10.1257/000282803321455214.

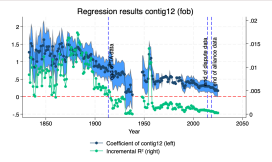
DATA (11)

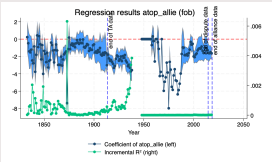
Control variables

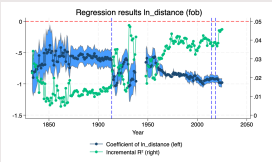
- Country x direction fixed effects
- distance (Centroids from Wikidata, not time varying)
- contiguity (Correlates of War + our own work)
- common empire (GeoPolHist up to 2022)
- alliance (inherited from sovereign) (Alliance Treaty Obligations and Provisions (ATOP) up to 2018)
- conflict (inherited from sovereign) (Militarized Interstate Disputes (MID) up to 2014)
- Trade agreements (Pahre (2008)'s data up to 1913)

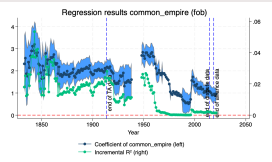

Regression results TA (fob)
end of TA data
Year
Coefficient of TA (left)
Incremental R² (right)


Regression results mid_n (fob)
start of TA data
end of TA data
end of distance data
Year
Coefficient of mid_n (left)
Incremental R² (right)

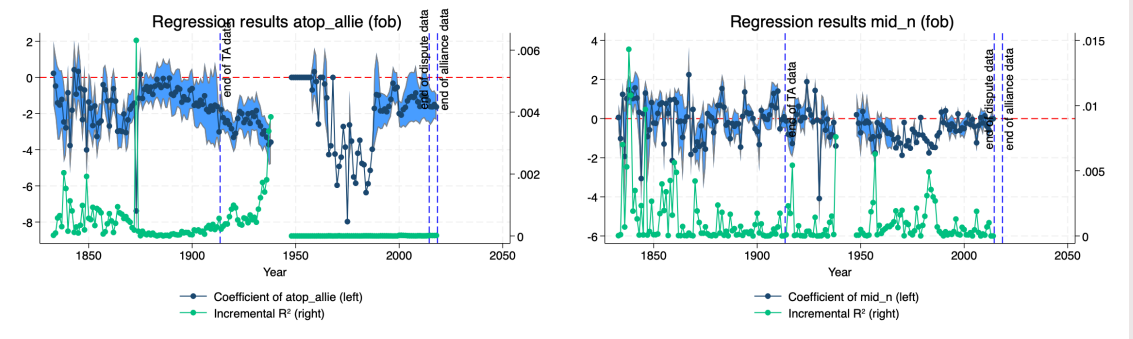

Regression results contig12 (fob)
start of TA data
end of TA data
end of distance data
Year
Coefficient of contig12 (left)
Incremental R² (right)

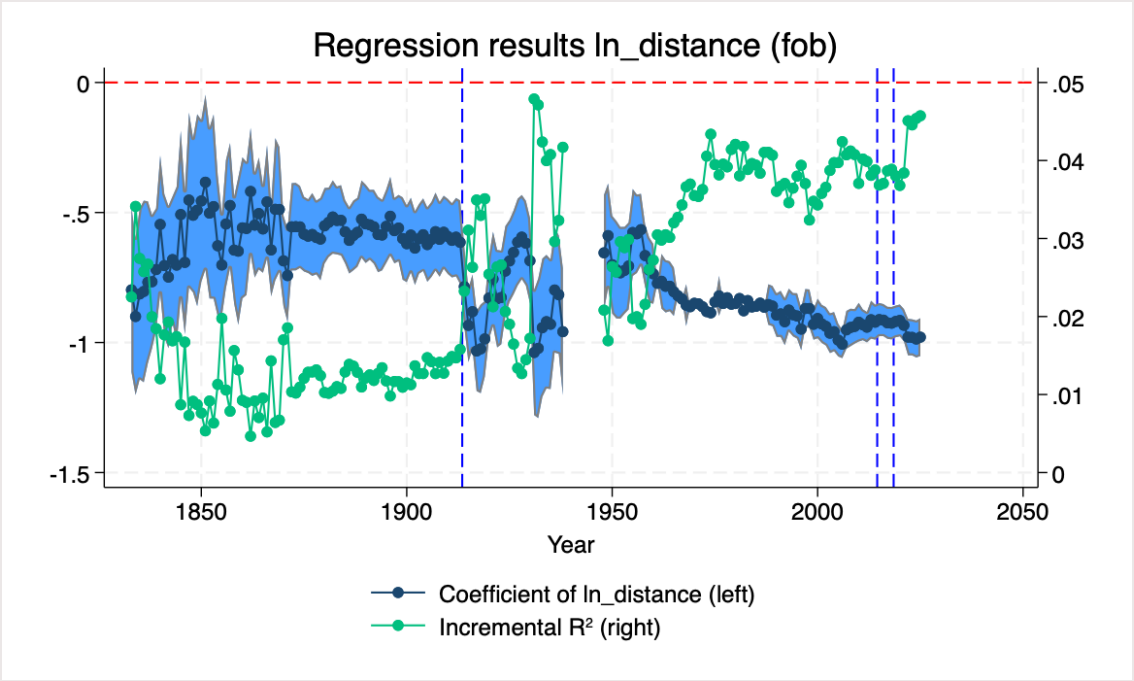

Regression results atop_allie (fob)
end of TA data
end of TA data
end of distance data
Year
Coefficient of atop_allie (left)
Incremental R² (right)

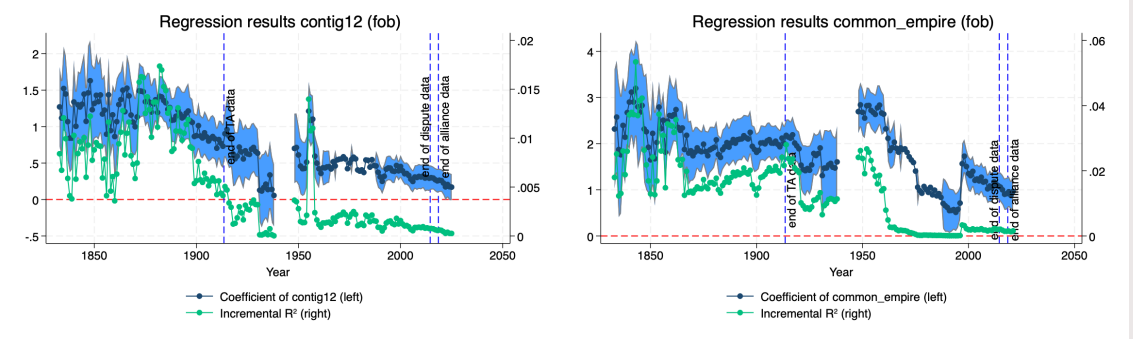

Regression results ln_distance (fob)
Year
Coefficient of ln_distance (left)
Incremental R² (right)


Regression results common_empire (fob)
end of TA data
end of TA data
end of distance data
Year
Coefficient of common_empire (left)
Incremental R² (right)

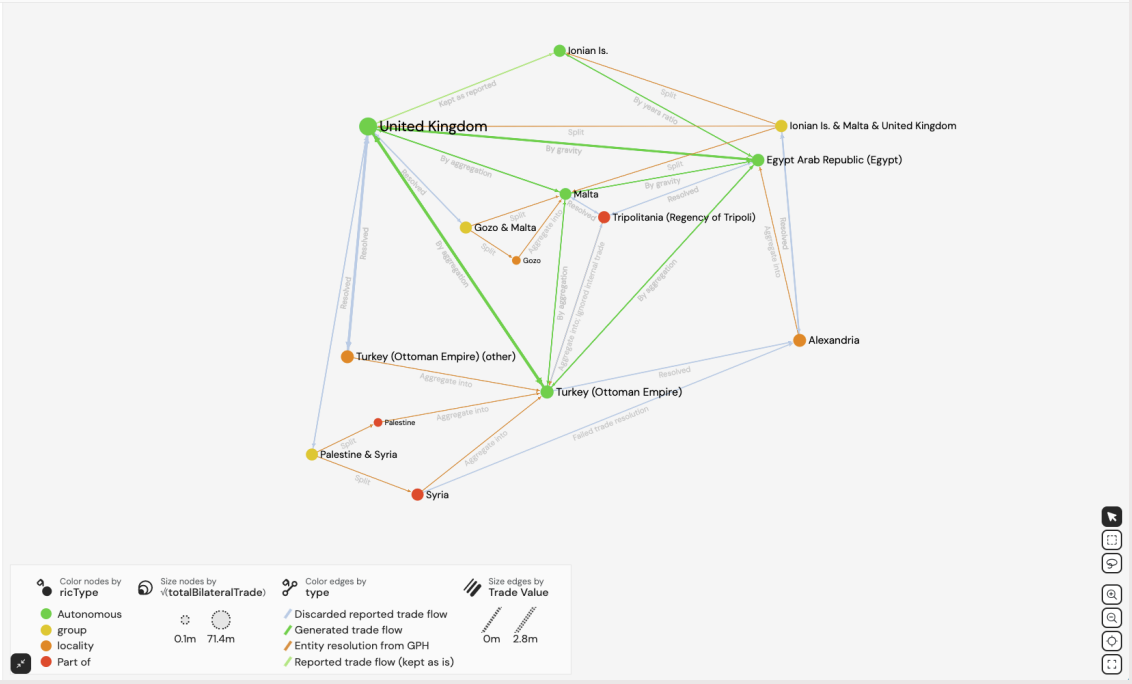

Regression results TA (fob)
2
1
0
-1
-2
.015
.01
.005
0
1840
1860
1880
1900
1920
Year
end of TA data
Coefficient of TA (left)
Incremental R² (right)


Regression results atop_allie (fob)
end of TA data
end of dispute data
end of alliance data
2
0
-2
-4
-6
-8
.006
.004
.002
0
1850
1900
1950
2000
2050
Year
Coefficient of atop_allie (left)
Incremental R² (right)
Regression results mid_n (fob)
end of TA data
end of dispute data
end of alliance data
4
2
0
-2
-4
-6
.015
.01
.005
0
1850
1900
1950
2000
2050
Year
Coefficient of mid_n (left)
Incremental R² (right)


Regression results ln_distance (fob)
0
-.5
-1
-1.5
.05
.04
.03
.02
.01
0
1850
1900
1950
2000
2050
Year
Coefficient of ln_distance (left)
Incremental R² (right)

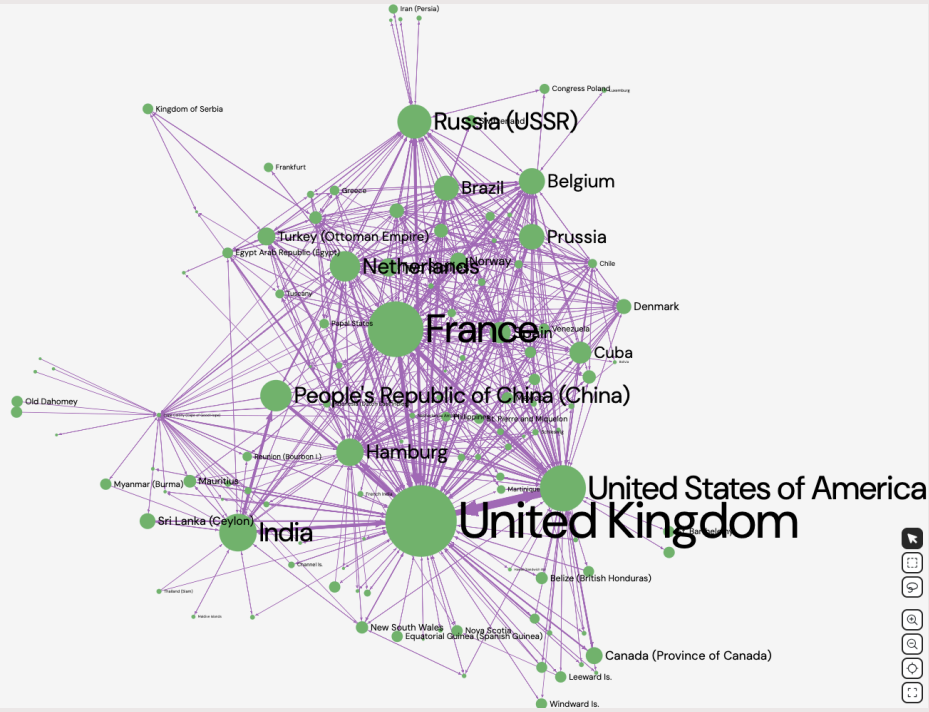

Regression results contig12 (fob)
2
1.5
1
.5
0
-.5
.02
.015
.01
.005
0
1850
1900
1950
2000
2050
Year
end of TA data
end of dispute data
end of alliance data
Coefficient of contig12 (left)
Incremental R² (right)
Regression results common_empire (fob)
4
3
2
1
0
.06
.04
.02
0
1850
1900
1950
2000
2050
Year
end of TA data
end of dispute data
end of alliance data
Coefficient of common_empire (left)
Incremental R² (right)

Examples of resolution around Malta, 1850

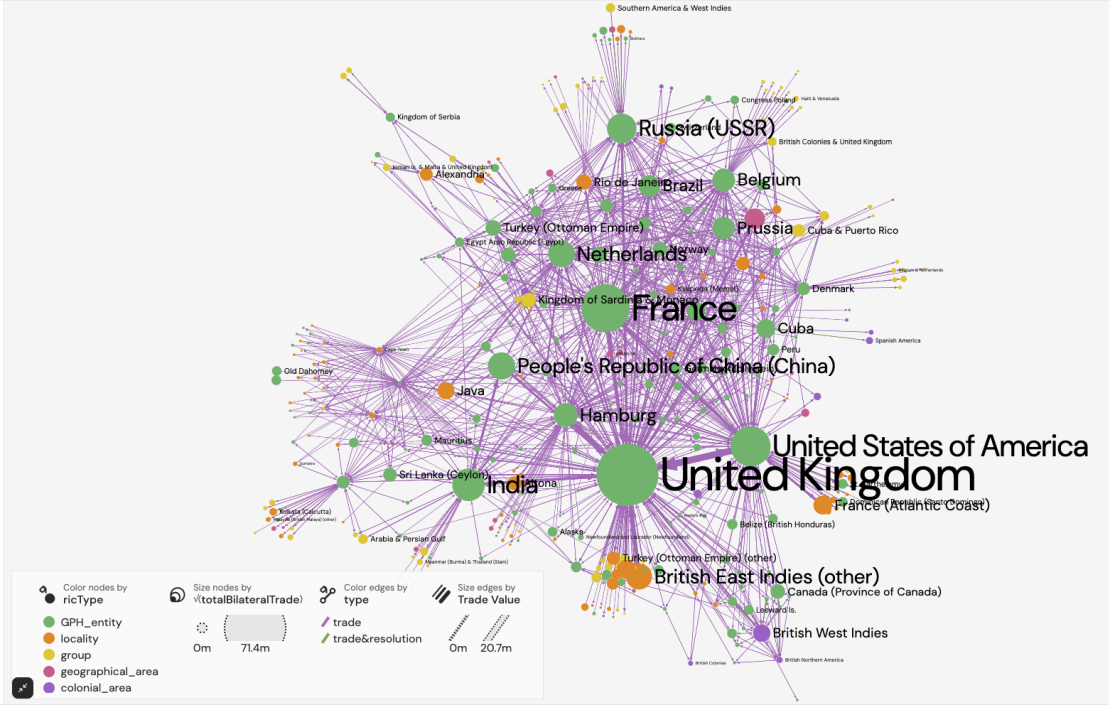


This is an extract around Malta in 1850 that illustrate the diversity of resolution

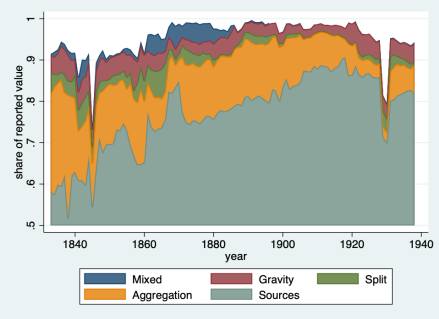
The 1850 trade network after the gravity step



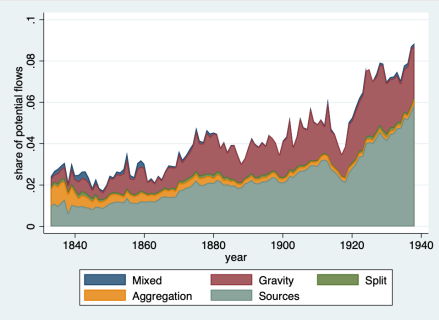
The 1850 trade network



How much trade do we normalise? Value



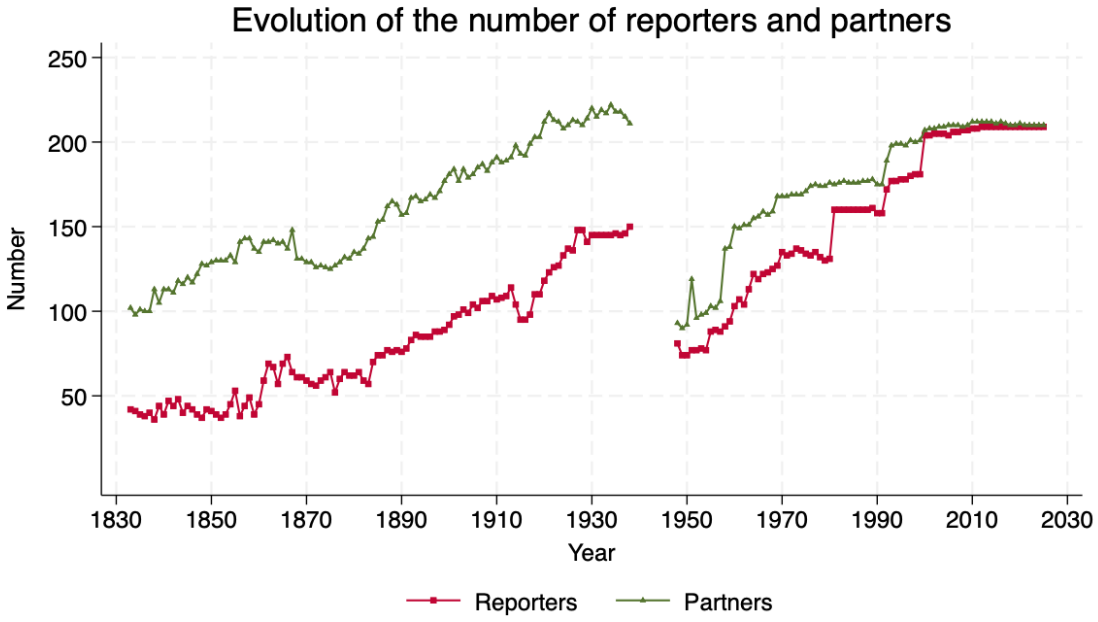
How much trade do we normalise? Flows

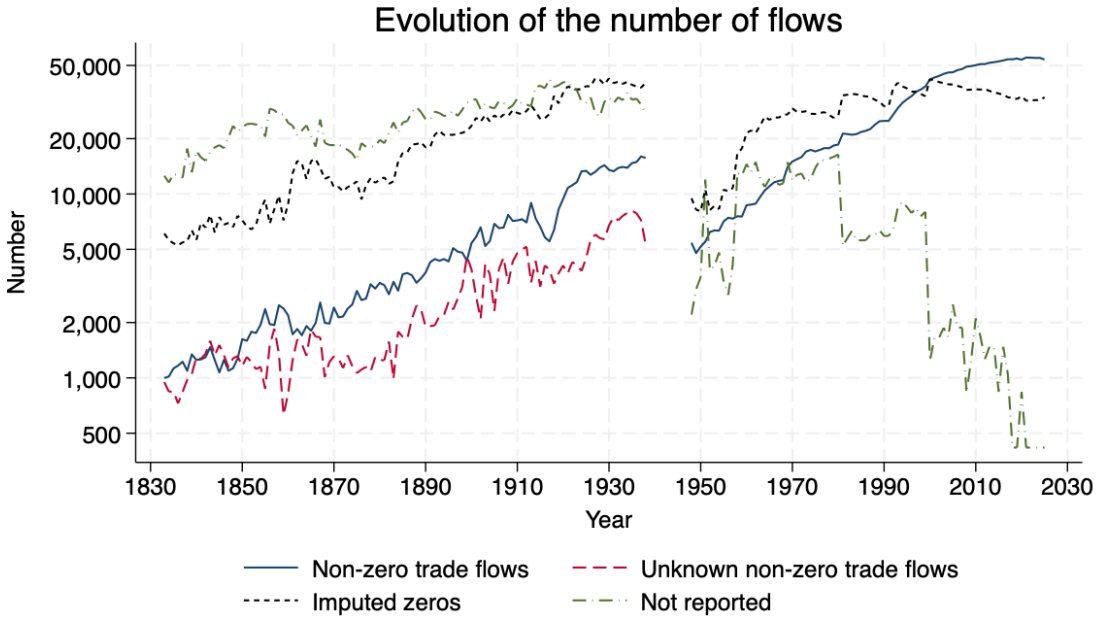


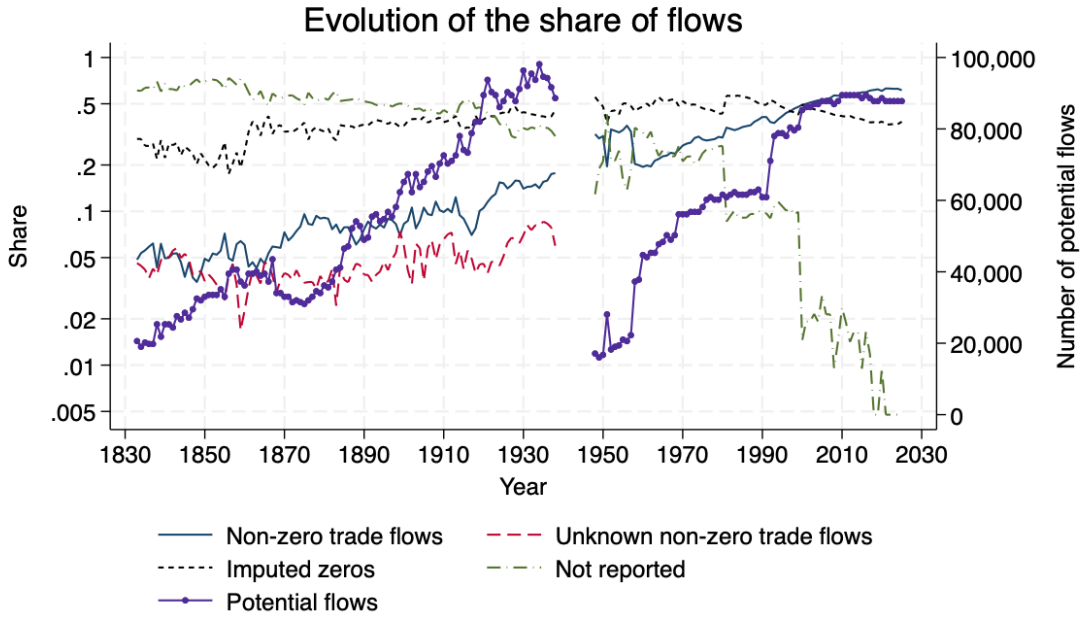
Keep it mind that we are working with the fringe of the trade networks, ou problematic flows are numerous but with smaller values on average

For the study of the extensive margin, flows are very important

The effect on the number of trade flows is much larger than on the value of trade flows


Evolution of the number of reporters and partners
Number
250
200
150
100
50
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2030
Year
Reporters
Partners


Evolution of the number of flows
Number
50,000
20,000
10,000
5,000
2,000
1,000
500
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2030
Year
Non-zero trade flows
Unknown non-zero trade flows
Imputed zeros
Not reported


Evolution of the share of flows
Share
1
.5
.2
.1
.05
.02
.01
.005
100,000
80,000
60,000
40,000
20,000
0
Number of potential flows
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2030
Year
Non-zero trade flows
Unknown non-zero trade flows
Imputed zeros
Not reported
Potential flows

II. IDENTIFYING TRADE COMMUNITIES

IDENTIFYING TRADE COMMUNITIES (1)

« The problem of community detection requires the partition of a network into communities of densely connected nodes, with the nodes belonging to different communities being only sparsely connected » Blondel et al. (2008)

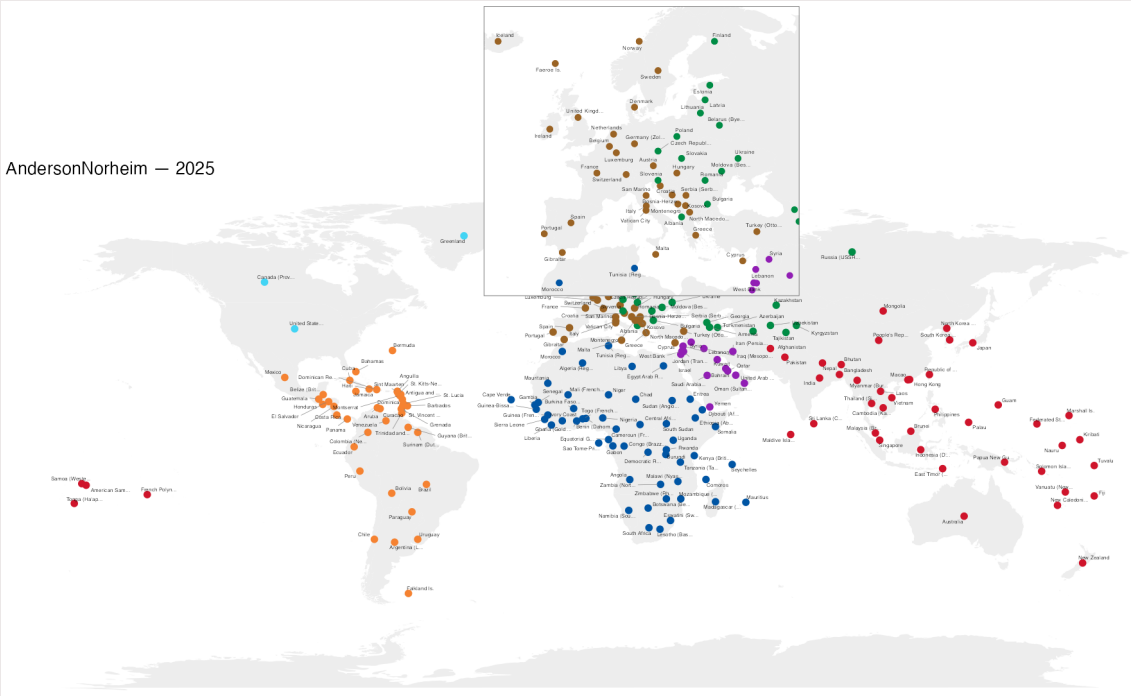
Trade studies: Either exogenous, sometimes endogenous, such as IntraMax

Poon (1997)

Network community has worked a lot on the subject

Blondel et al. (2008) : Louvain

Anderson and Norheim (1993)'s blocs as a reference



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Western Europe (including Turkey and Yugoslavia)
Eastern Europe (East of West Germany, including Soviet Central Asia)

North America (excluding Mexico and West Indies)
Latin America

(Japan)
Australasia
Developing Asia

Africa
Middle East (as far as Iran)

Anderson, Kym, and Hege Norheim. "Is world trade becoming more regionalized?." *Review of International Economics*, 1.2 (1993): 91-109.

Finger, Karl-Michael, Hege Norheim, and Kym Anderson. 1993. "Trends in the Regionalization of World Trade, 1928 to 1990." In *Regional Integration and the Global Trading System*, by Kim Anderson and Richard Blackhurst. https://www.cabidigitallibrary.org/doi/full/10.5555/19941801183.

This is the most common point of view. Anderson and Norheim. Currency blocs, political trade communities, empires...

This leads to an endogeneity issue (you might pick the composition of a bloc because you see increased inbloc trade).

It also leads to fixity over time. The relevant bloc might actually change through time, and you might miss that. Most obvious example : maybe imperial blocs are important in the 1930s and 1950s, but they loose that importance through time. And now continental blocs are important because of regional trade agreements, especially the EU.

IDENTIFYING TRADE COMMUNITIES (3)

Two issues:

1. What is the algorithm?
2. What is the weight/proximity mesure?

IDENTIFYING TRADE COMMUNITIES (4) ALGORITHMS

Intramax algorithm:

1. Computes all proximities between actors/groups
2. Regroup the closest pair
3. Repeat till the regrouping increases internal trade in the group by more than 10% of world trade

Louvain algorithm:

Louvain adds a global measure grouping quality "modularity" (Newman (2006))

$$Q = \frac{1}{2m} . \sum_{i,j} \left(A_{i,j} . \frac{k_i . k_j}{2m} . \delta(c_i, c_j) \right)$$

$$Q \in [-0.5, 1]$$

Which is the ratio between the weight of links inside groups compared to random links

M. E. J. Newman, « Modularity and community structure in networks », *Proc. Natl. Acad. Sci. USA*, vol. 103, no 23, 2006, p. 8577–8582 https://dx.doi.org/10.1073%2Fpnas.0601602103

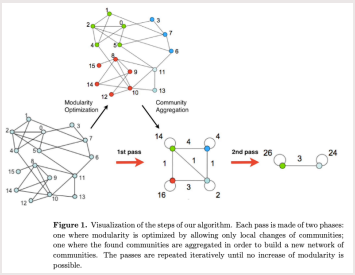


Figure 1. Visualization of the steps of our algorithm. Each pass is made of two phases: one where modularity is optimized by allowing only local changes of communities; one where the found communities are aggregated in order to build a new network of communities. The passes are repeated iteratively until no increase of modularity is possible.

$A_{i,j}$ is the weight of the link (or proximity) between polities i and j that we just defined

$k_i = \sum_i A_{i,j}$ is total weight of a polity's links (it might be different from one)

$m = \frac{1}{2} . \sum_{i,j} A_{i,j}$ is the sum of all weights in the trade network divided by two

c_i and c_j are the blocs of polities i and j. The δ-function $\delta(u, v)$ is 1 if $u = v$ and 0 otherwise

All polities are put in their own bloc
We examine all trade partners j of a polity i and we assign i in j's bloc, where j's bloc provides the largest gain in modularity. If there are no positive gains, i stays in its bloc. Once all polities have been considered once, we examine them again in the same order and they can moved from bloc to bloc.
When all positive gain-grouping have been made, we re-create the network with the new blocs as units, considering internal trade as a self-loop. Proximity between blocs is the sum of proximity of each pair of polities. And we iterate (2 and 3) till no change in grouping is made in phase 2.
The result depends on the order in which polities are examined. We run it multiple time and aggregate the results.

Blondel, Vincent D., Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. 2008. "Fast Unfolding of Communities in Large Networks." *Journal of Statistical Mechanics: Theory and Experiment* 2008 (10): P10008.

Pougué-Biyong, John N., and Renaud Lambiotte. "SignedLouvain: Louvain for signed networks." arXiv preprint arXiv:2407.19288 (2024).

IDENTIFYING TRADE COMMUNITIES (5) WEIGHT/PROXIMITY

We can have up to four flows by pair.

But we cannot simply look at exports or imports because of the large number of missing observations

In all cases, we compute 1 to 4 proximities and we take the max

$$Max\left(P_{ij}^{X};P_{ji}^{X};P_{ij}^{M};P_{ji}^{M}\right)=A_{ij}=A_{ji}$$

1. We can just take raw trade

2. Intramax proximity

One looks for the polity/bloc pair that is the furthest from a random model of bilateral trade.
X_{ij} is (eg) fob exports from polity/bloc i to polity/bloc j
X'_{ij} is counterfactual, random, no friction trade, defined as:

$$X'_{ij}=\frac{X_{i.}}{X}\frac{X_{.j}}{X}X=\frac{X_{i.}X_{.j}}{X}$$

Weight/Proximity: $(X_{ij}-X'_{ij})+(X_{ji}-X'_{ji})[+(M_{ij}-M'_{ij})+(M_{ji}-M'_{ji})]$

X is world exports, Xi. is polity/bloc i's exports. X.j is exports going to polity/bloc j.

- Poon's maximisation objective is in monetary terms, and as such gives an advantage to large polity pairs.
  - We believe if a small polity does 100% of its trade with a big one, even if this trade is small in monetary terms, they must be in the same bloc. So we normalise. Kohl & Brouwer (2014)

    Proximity $= \frac{X_{ij} - X'_{ij}}{X_{i.}}, \frac{X_{ji} - X'_{ji}}{X_{j.}}, \frac{M_{ij} - M'_{ij}}{M_{.j}} and \frac{M_{ji} - M'_{ji}}{M_{.I}}$

3. TIBI (Trade Intensity Bilateral Index)

Poon's maximisation objective can also be expressed as a ratio : CEPII's Relative bilateral intensity of port Intensity:

$$X_{ij}/X'_{ij} = (X_{ij}/X_{i.})/(X_{.j}/X)) \in [-1, X/X_{.j}].$$

Inspired by Balassa (1965)'s index of comparative advantage. Eg Gaulier and Ünal-Kesenci (2004)

Iapadre (2006), Daudin et al. (2011) mention three issues:

Solving range variability

Solution: $I^1_{ij} = (X_{ij}/X_{i.})/(X_{rj}/X)) \in [0, +\infty[$

Solving range asymmetry

Solution: $I^2_{ij} = (I^1_{ij} - 1)/(I^1_{ij} + 1) \in [-1, 1]$

Solving dynamic ambiguity

Solution: $TIBI_{ij} = (I^1_{ij}/I^3_{ij} - 1)/(I^1_{ij}/I^3_{ij} + 1) \in [-1, 1]$

Where: $I^3 = (1 - (X_{ij}/X_{i.}))/(1 - (X_{rj}/X))$

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CEPII's "Relative bilateral intensity of exports" Deutsch and Savage (1960), Drysdale and Garnet (1982), Freudenberg, Gaulier and Ûnal-Kensenci (1998), pp.66-68

The maximum value of the index depends on the size of $X_{.j}

The range is not symmetric around neutrality. I1 ranges from 0 to 1 if trade is less important than expected and from 1 to infinity if trade is more important than expected. To solve that issue, we use the following index, which varies between -1 and 1 and has neutrality at 0 rather than 1:

A change in the index might be in the same direction as the change in the complementary index measuring the intensity of trade between region i and all non-j regions.

Daudin, Guillaume, Christine Rifflart, and Danielle Schweisguth. 2011. "Who Produces for Whom in the World Economy?" *Canadian Journal of Economics/Revue Canadienne d'économique* 44 (4): 1403–37. https://doi.org/10.1111/j.1540-5982.2011.01679.x.

Iapadre, Lelio. 2006. "Regional Integration Agreements and the Geography of World Trade." *Assessment And Measurement of Regional Integration*, 65–85.

Proximity measures in 1938

expCountry	impCountry	Poon1	Poon2	REI	I1	I2	I3	TIBI
United Kingdom	India	27,102,807	0.24	5.07	19.25	0.90	0.71	0.93
Canada (Province of Canada)	United States of America	54,593,186	0.31	2.73	6.05	0.72	0.56	0.83
United States of America	Canada (Province of Canada)	49,856,880	0.19	2.16	6.33	0.73	0.69	0.80
France	Belgium	18,704,330	0.29	4.18	6.05	0.72	0.66	0.80
Netherlands	Belgium	7,244,770	0.14	2.58	2.93	0.49	0.83	0.56
India	United Kingdom	17,237,537	0.27	1.71	1.83	0.29	0.55	0.54
United Kingdom	Netherlands	5,970,202	0.05	1.83	2.16	0.37	0.93	0.40
France	Netherlands	3,799,255	0.06	1.92	2.11	0.36	0.93	0.39
United Kingdom	France	6,673,710	0.06	1.79	2.09	0.35	0.93	0.39
Netherlands	France	2,988,465	0.06	1.79	1.90	0.31	0.93	0.34
Netherlands	United Kingdom	7,292,692	0.14	1.38	1.42	0.17	0.76	0.31
United States of America	France	7,103,810	0.03	1.36	1.70	0.26	0.96	0.28
Canada (Province of Canada)	United Kingdom	16,696,877	0.09	1.25	1.36	0.15	0.81	0.25
United Kingdom	Canada (Province of Canada)	4,344,973	0.04	1.24	1.29	0.13	0.95	0.16
United States of America	Netherlands	2,763,814	0.01	1.16	1.28	0.12	0.98	0.13
United States of America	United Kingdom	4,983,666	0.02	1.05	1.08	0.04	0.95	0.06
United Kingdom	United States of America	392,914	0.00	1.02	1.02	0.01	1.00	0.01
India	France	-201,969	0.00	0.96	0.95	-0.02	1.00	-0.03
India	United States of America	-1,106,690	-0.02	0.90	0.90	-0.06	1.02	-0.07
France	United States of America	-1,655,569	-0.03	0.86	0.85	-0.08	1.03	-0.10
India	Netherlands	-797,324	-0.01	0.81	0.79	-0.12	1.01	-0.12
United Kingdom	Belgium	-1,986,071	-0.02	0.81	0.78	-0.12	1.02	-0.14
France	United Kingdom	-3,712,237	-0.06	0.85	0.84	-0.09	1.10	-0.14
India	Belgium	-2,512,095	-0.04	0.57	0.55	-0.29	1.05	-0.31
United States of America	Belgium	-8,424,108	-0.03	0.65	0.54	-0.30	1.06	-0.32
Netherlands	United States of America	-4,822,450	-0.10	0.46	0.45	-0.38	1.13	-0.43
United States of America	India	-8,882,670	-0.03	0.43	0.33	-0.50	1.06	-0.52
Netherlands	India	-2,077,787	-0.04	0.30	0.29	-0.55	1.05	-0.57
France	India	-2,776,845	-0.04	0.28	0.26	-0.59	1.05	-0.61
Canada (Province of Canada)	Netherlands	-8,525,517	-0.05	0.24	0.19	-0.68	1.07	-0.69
Canada (Province of Canada)	Belgium	-13,026,826	-0.07	0.19	0.15	-0.74	1.11	-0.77
India	Canada (Province of Canada)	-8,806,570	-0.14	0.15	0.14	-0.75	1.18	-0.78
Canada (Province of Canada)	France	-11,695,530	-0.07	0.12	0.09	-0.83	1.10	-0.85
France	Canada (Province of Canada)	-9,490,448	-0.15	0.09	0.09	-0.84	1.19	-0.87
Netherlands	Canada (Province of Canada)	-7,415,260	-0.15	0.09	0.08	-0.84	1.19	-0.87
Canada (Province of Canada)	India	-9,552,616	-0.05	0.08	0.07	-0.88	1.08	-0.88

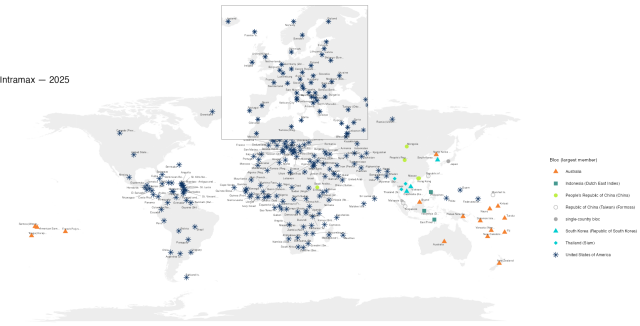
Identifying trade communities (6)

1. Intramax with Poon 2 measure ($\frac{X_{ij}-X'_{ij}}{X_{i.}}$)
2. Louvain (only accepts positive weights)

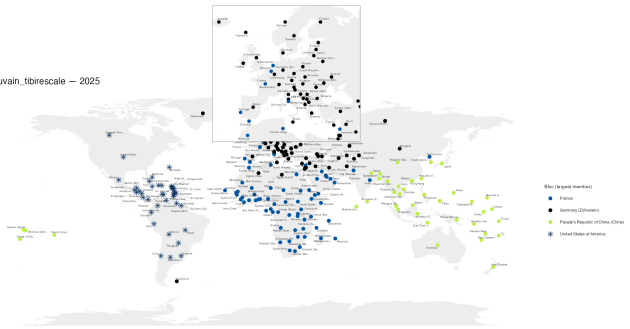
   a. With raw trade

   b. With TIBI + 1
   Issue : missing and zeros are treated the same way
3. Signed Louvain with raw TIBI

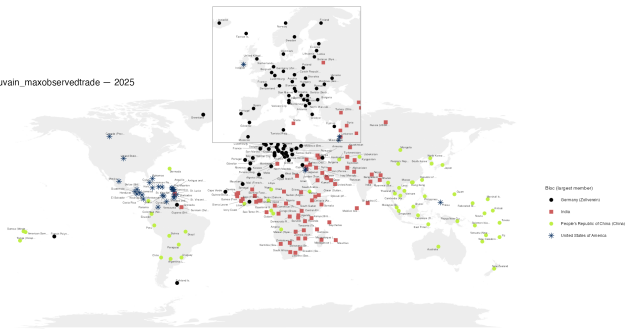
Intramax — 2025



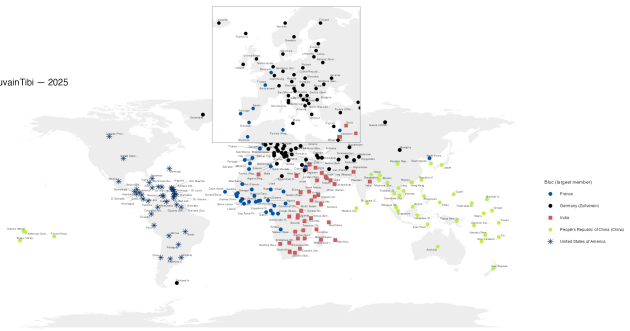
Louvain_tibirescale — 2025

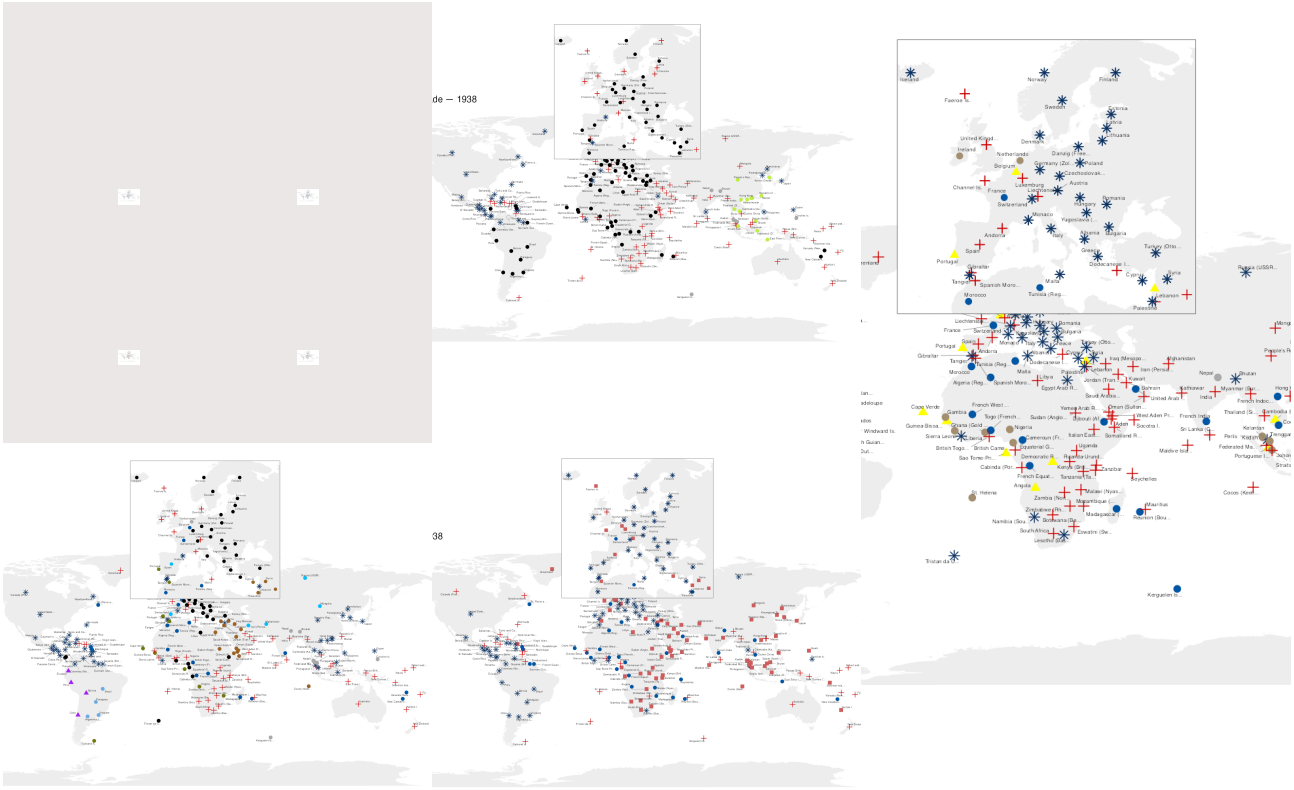


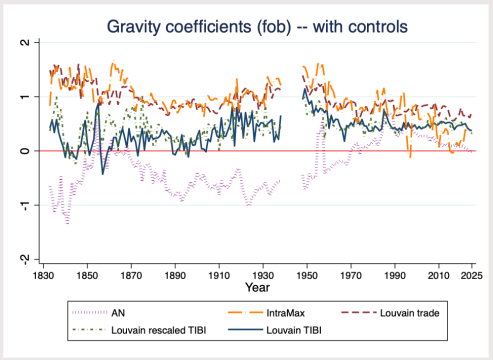
Louvain_maxobservedtrade — 2025

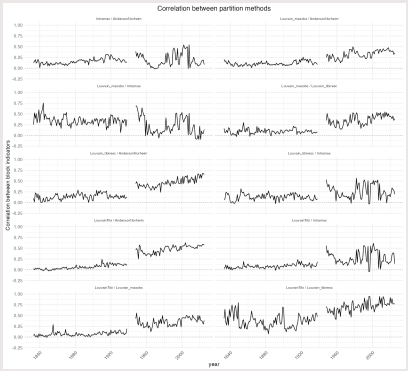


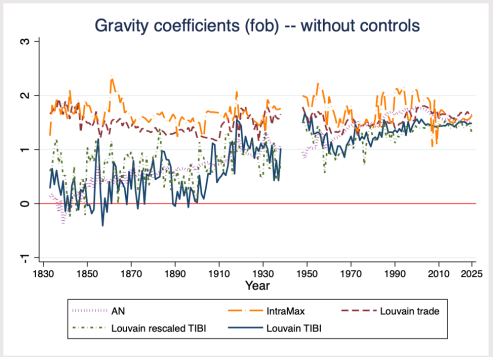
LouvainTibi — 2025

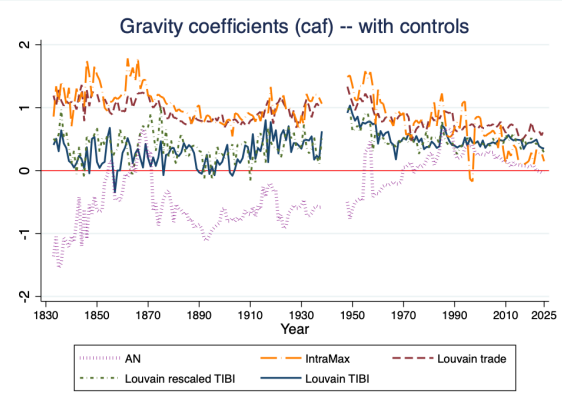


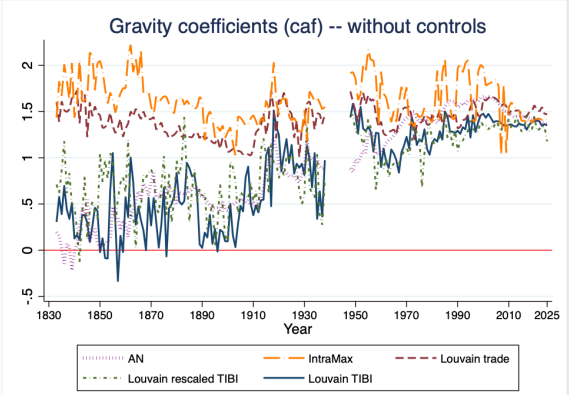

de — 1938
38

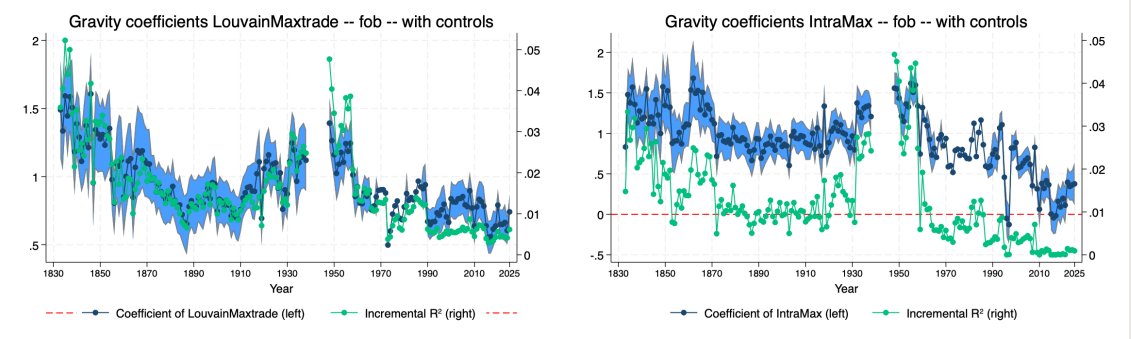

Gravity coefficients (fob) -- with controls
2
1
0
-1
-2
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
AN
IntraMax
Louvain trade
Louvain rescaled TIBI
Louvain TIBI


Correlation between partition methods
Correlation between block indicators
year


Gravity coefficients (fob) -- without controls
3
2
1
0
-1
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
AN
IntraMax
Louvain trade
Louvain rescaled TIBI
Louvain TIBI


Gravity coefficients (caf) -- with controls
2
1
0
-1
-2
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
AN
IntraMax
Louvain trade
Louvain rescaled TIBI
Louvain TIBI


Gravity coefficients (caf) -- without controls
2
1.5
1
.5
0
-.5
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
AN
IntraMax
Louvain trade
Louvain rescaled TIBI
Louvain TIBI

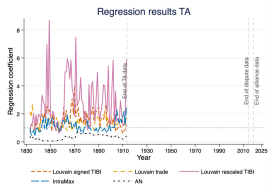

Gravity coefficients LouvainMaxtrade -- fob -- with controls
2
1.5
1
.5
.05
.04
.03
.02
.01
0
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
Coefficient of LouvainMaxtrade (left)
Incremental R² (right)
Gravity coefficients IntraMax -- fob -- with controls
2
1.5
1
.5
0
-.5
.05
.04
.03
.02
.01
0
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
Coefficient of IntraMax (left)
Incremental R² (right)

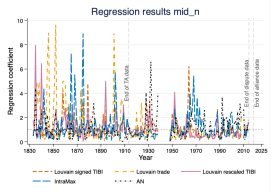
IV. DETERMINANTS OF TRADE COMMUNITIES

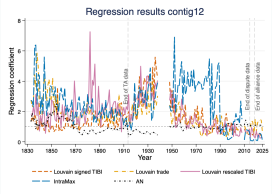
Logit regression on polity-pairs each year, with the usual control variables

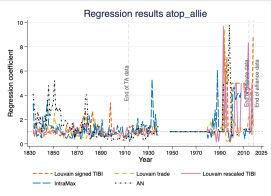
No causality, obviously\

We compute the incremental R^2 (but we need a better idea)

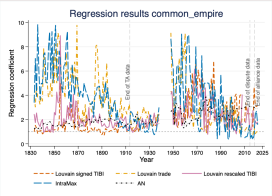

Regression results TA
Regression coefficient
Year
End of TA data
End of dispute data
End of alliance data
Louvain signed TIBI
Louvain trade
Louvain rescaled TIBI
IntraMax
AN

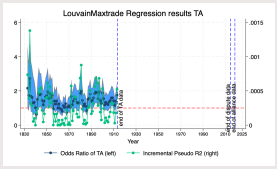

Regression results mid_n
Regression coefficient
Year
End of TA data
End of dispute data
End of alliance data
Louvain signed TIBI
Louvain trade
Louvain rescaled TIBI
IntraMax
AN

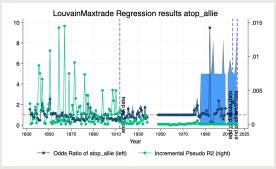

Regression results contig12
Regression coefficient
Year
End of TA data
End of dispute data
End of alliance data
Louvain signed TIBI
Louvain trade
Louvain rescaled TIBI
IntraMax
AN

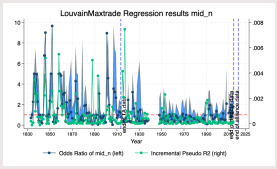

Regression results atop_allie
Regression coefficient
Year
End of TA data
End of dispute data
End of alliance data
Louvain signed TIBI
Louvain trade
Louvain rescaled TIBI
IntraMax
AN

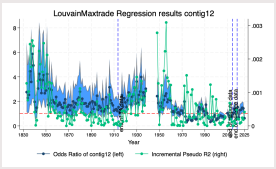

Regression results ln_distance
Regression coefficient
Year
End of TA data
End of dispute data
End of alliance data
Louvain signed TIBI
Louvain trade
Louvain rescaled TIBI
IntraMax
AN

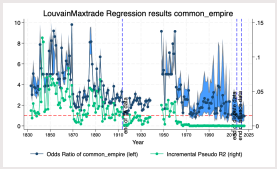

Regression results common_empire
Regression coefficient
Year
End of TA data
End of dispute data
End of alliance data
Louvain signed TIBI
Louvain trade
Louvain rescaled TIBI
IntraMax
AN

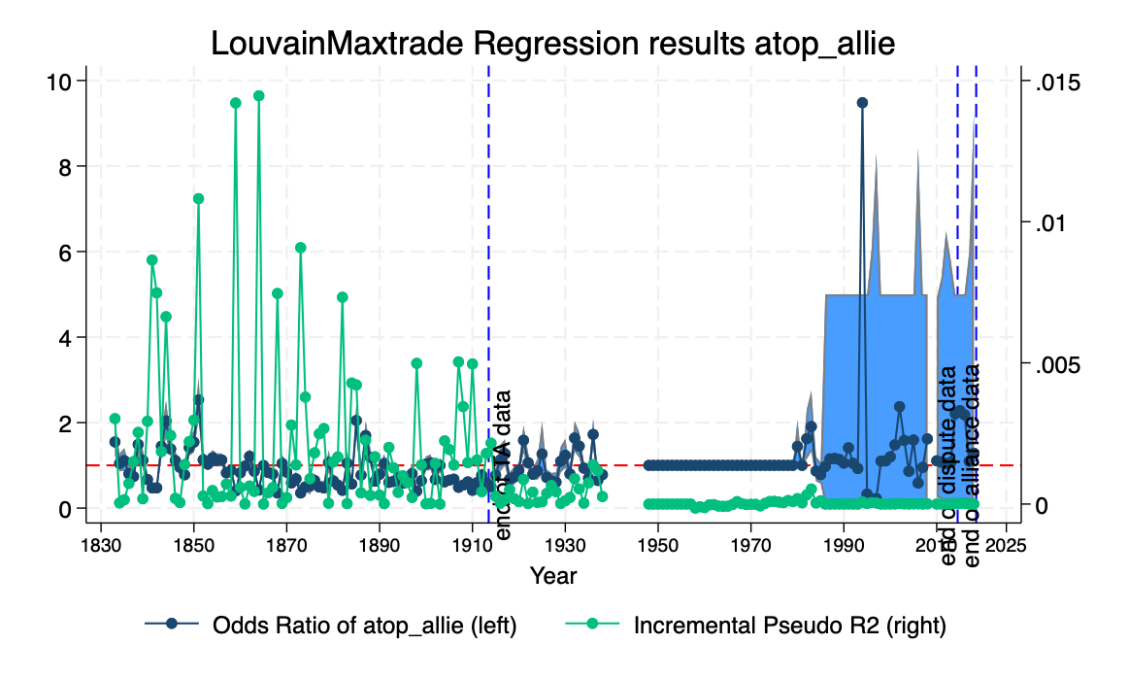

LouvainMaxtrade Regression results TA
Year
Odds Ratio of TA (left)
Incremental Pseudo R2 (right)

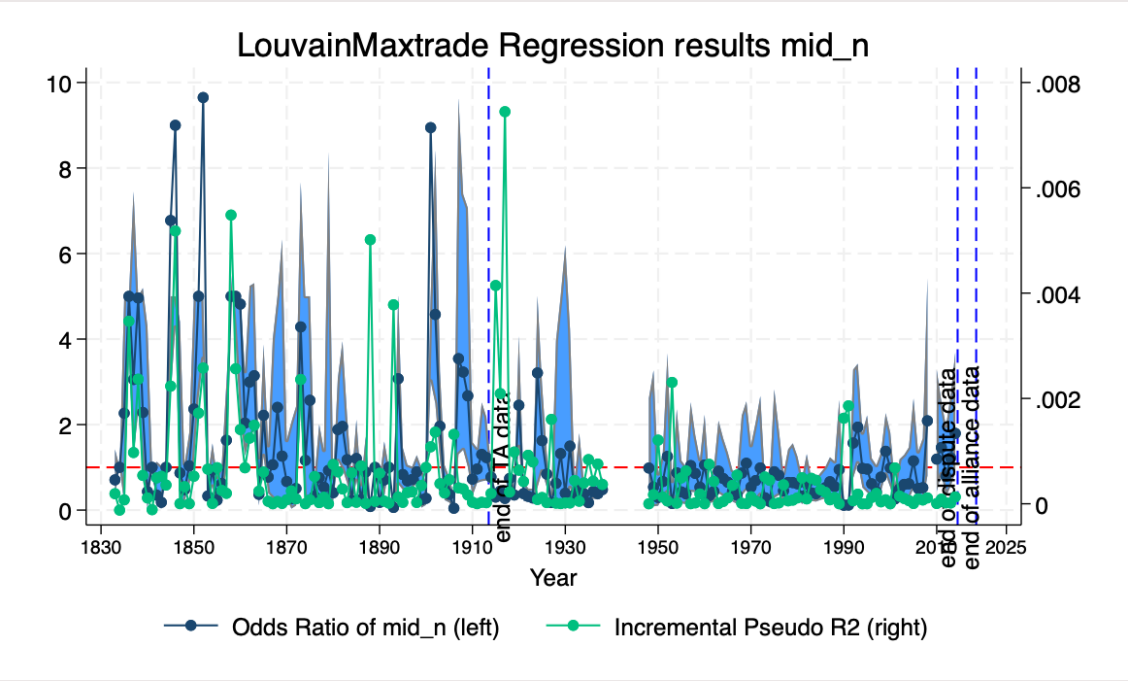

LouvainMaxtrade Regression results atop_allie
Year
Odds Ratio of atop_allie (left)
Incremental Pseudo R2 (right)

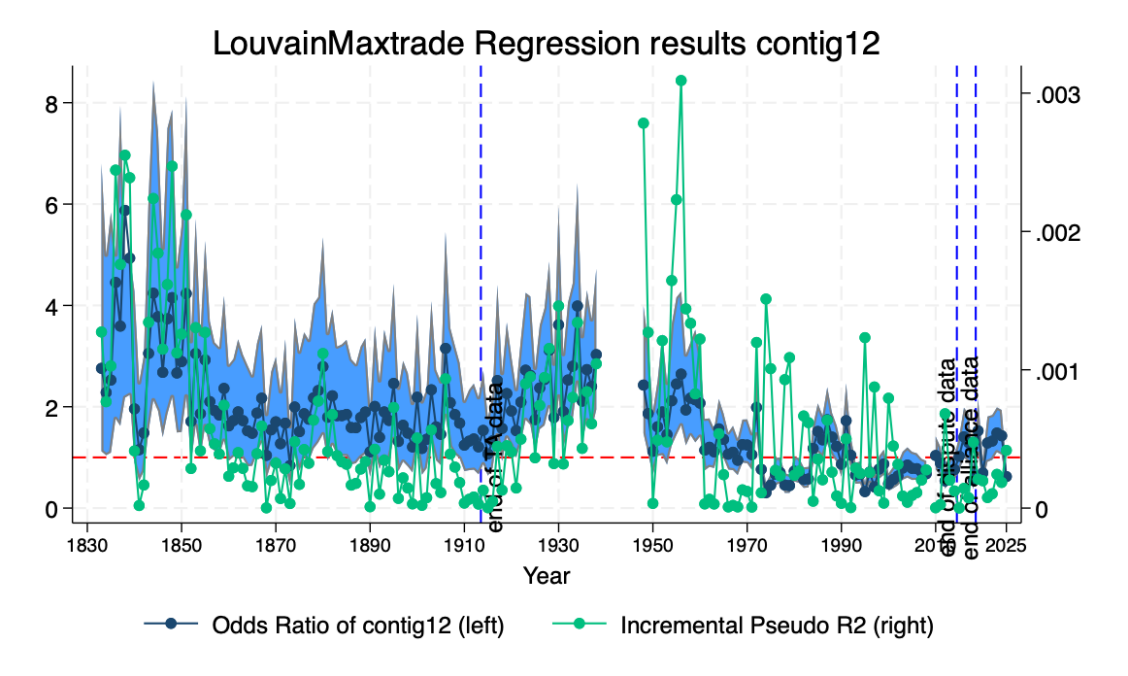

LouvainMaxtrade Regression results mid_n
Year
Odds Ratio of mid_n (left)
Incremental Pseudo R2 (right)

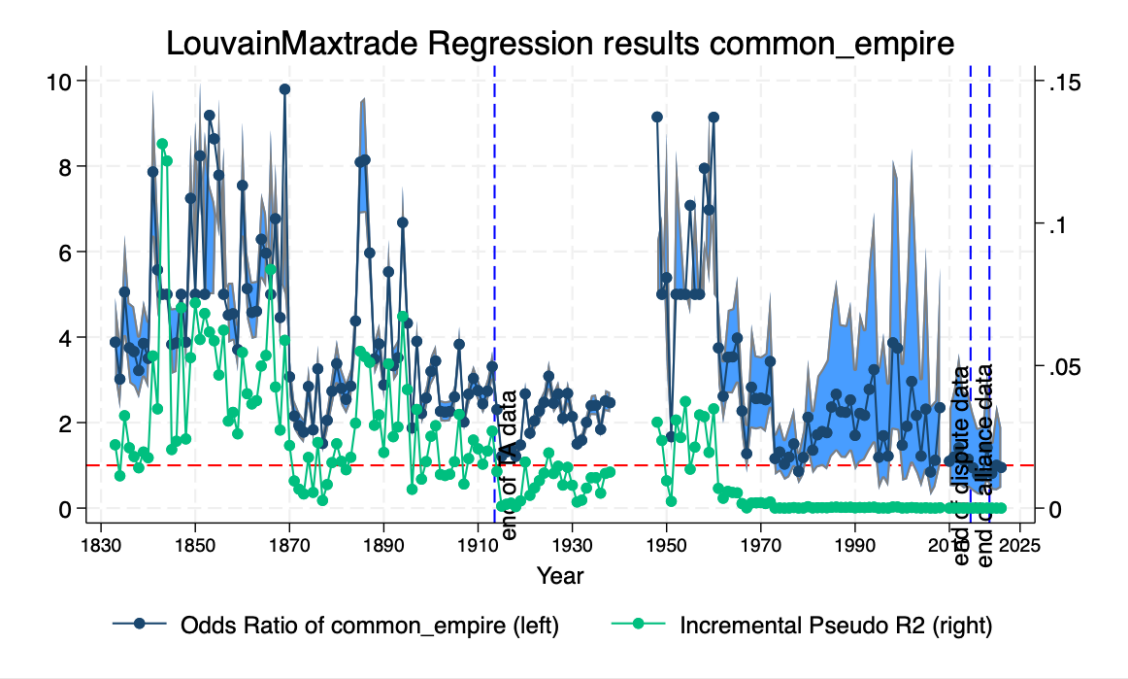

LouvainMaxtrade Regression results contig12
Year
Odds Ratio of contig12 (left)
Incremental Pseudo R2 (right)

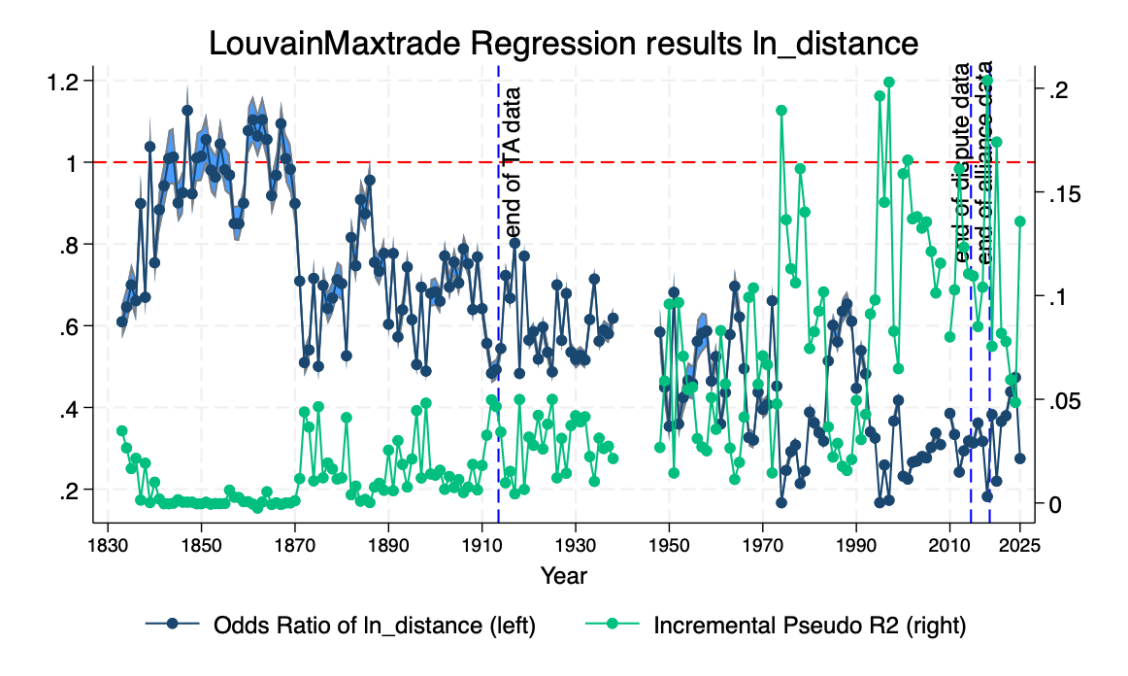

LouvainMaxtrade Regression results common_empire
Year
Odds Ratio of common_empire (left)
Incremental Pseudo R2 (right)


LouvainMaxtrade Regression results atop_allie
10
8
6
4
2
0
.015
.01
.005
0
1830
1850
1870
1890
1910
1930
1950
1970
1990
2025
Year
end of MA data
end of dispute data
end of alliance data
Odds Ratio of atop_allie (left)
Incremental Pseudo R2 (right)


LouvainMaxtrade Regression results mid_n
10
8
6
4
2
0
.008
.006
.004
.002
0
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
end of dispute data
end of alliance data
Odds Ratio of mid_n (left)
Incremental Pseudo R2 (right)


LouvainMaxtrade Regression results contig12
8
6
4
2
0
.003
.002
.001
0
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
end of TA data
end of dispute data
end of alliance data
Odds Ratio of contig12 (left)
Incremental Pseudo R2 (right)


LouvainMaxtrade Regression results common_empire
10
8
6
4
2
0
.15
.1
.05
0
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
end of IA data
end of dispute data
end of alliance data
Odds Ratio of common_empire (left)
Incremental Pseudo R2 (right)


LouvainMaxtrade Regression results ln_distance
1.2
1
.8
.6
.4
.2
.2
.15
.1
.05
0
end of TA data
end of dispute data
end of alliance data
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
Odds Ratio of ln_distance (left)
Incremental Pseudo R2 (right)

IV. EFFECTS OF TRADE COMMUNITIES

Effects of trade communities (1)

Importance of the extensive margin (Helpman, Melitz & Rubinstein (2008), Hummels & Klenow (2005) and Felbermayr & Kohler (2006))

Basic model in the trade duration literature: Cox model with proportional hazard (Besedeš and Prusa (2006))

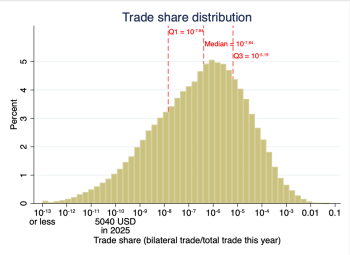
Method : cloglog. Said to be preferable to Cox model in discrete time (Hess and Persson (2012)

Or even log (weakly proportional hazards) or probit?

"Proportional hazards" means that covariates multiply a shared baseline risk by a constant factor, so the ratio of risks between any two individuals never changes over time.

Besedeš, Tibor, and Thomas J. Prusa. 2006. "Product Differentiation and Duration of US Import Trade." *Journal of International Economics* 70 (2): 339–58.

Hess, Wolfgang, and Maria Persson. 2012. "The Duration of Trade Revisited: Continuous-Time versus Discrete-Time Hazards." *Empirical Economics* 43 (3): 1083–107.



We have an issue with the gravity model as it creates very small non-zero trade flows. So we probably want to cut off very small trade flows -- we choose one tenth of a billionth (c. 5k $ in 2025). Same question with failed splits

Trading flows

Excluding ↓ / Including very small flows and failed splits →	0	1	Total
0	7,867,646	158,949	8,026,595
1		2,613,141	2,613,141
Total	7,867,646	2,772,090	10,639,736

Trading pairs

Excluding ↓ / Including very small flows and failed splits →	0	1	Total
0	89,212	89,235	178,447
1		92,981	92,981
Total	89,212	182,216	271,428

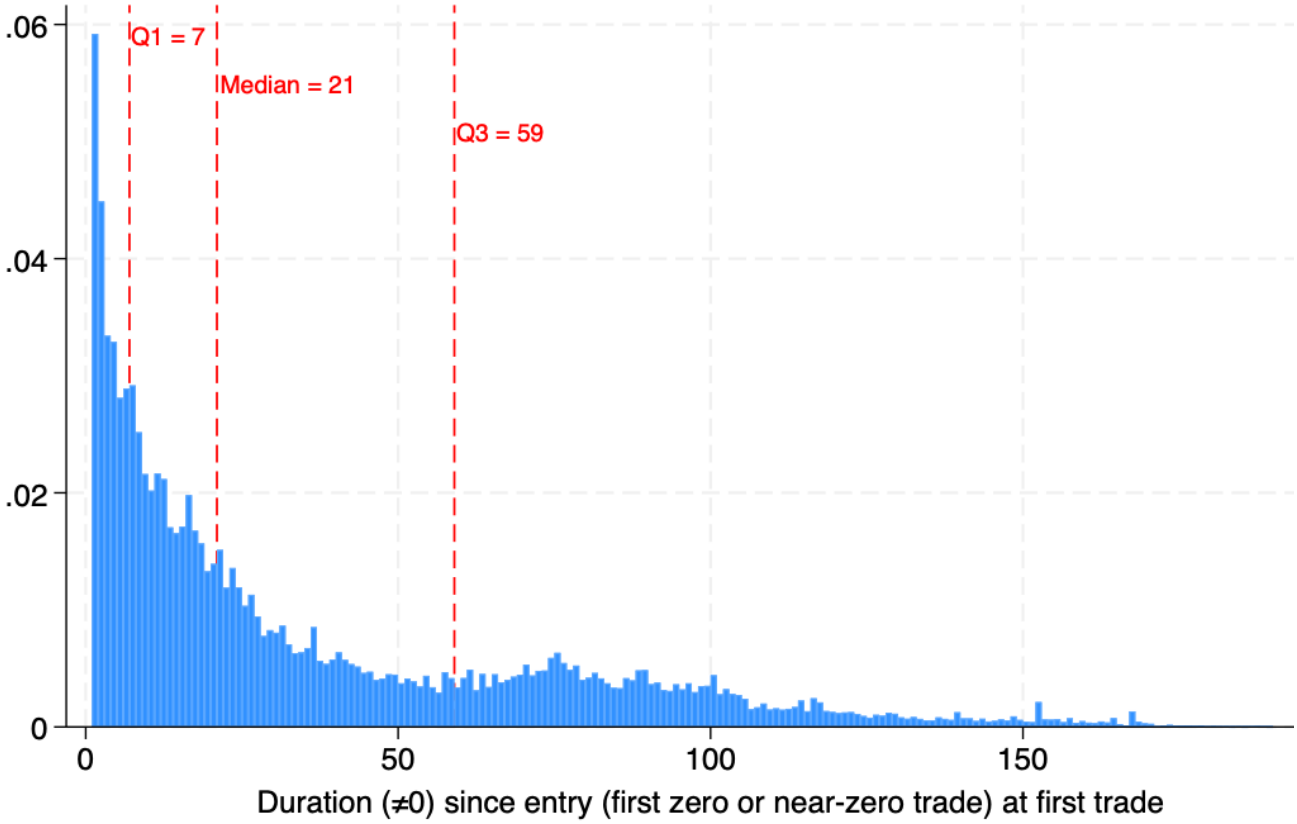
TRADING FLOWS "AT RISK" (EXCLUDING SECOND AND SUBSEQUENT TRADE OBSERVATIONS)

Excluding very small flows and failed splits	Total
0	6,387,135
1	86,158
Total	6,473,293

Including very small flows and failed splits	Total
0	6,220,999
1	91,083
Total	6,312,082

Time at risk

- Duration 1 : years of directed pair existence before first trade (not counting gaps)
- Duration 2 : time since appearance (counting gaps)
- Duration 3 : time since appearance with zero (and/or very small) trade



Some pairs stay at risk a long time !

Community definition : LouvainMaxtrade (cloglog, year fixed effects) -- Hazard ratios

	Start trade (excluding very small flows and failed splits)			Start trade (including very small flows and failed splits)		
	Time since appearance with zero trade	Years of existence	Time since appearance	Time since appearance with zero trade	Years of existence	Time since appearance
	LouvainMaxtrade	LouvainMaxtrade	LouvainMaxtrade	LouvainMaxtrade	LouvainMaxtrade	LouvainMaxtrade
Duration (years)	0.9976***	0.9986***	0.9961***	0.9976***	0.9986***	0.9962***
	[0.9970, 0.9982]	[0.9979, 0.9994]	[0.9956, 0.9967]	[0.9970, 0.9982]	[0.9978, 0.9993]	[0.9956, 0.9967]
Duration²	1.000042***	1.000051***	1.000043***	1.000041***	1.000048***	1.000043***
	[1.000037, 1.000047]	[1.000045, 1.000057]	[1.000040, 1.000047]	[1.000037, 1.000046]	[1.000042, 1.000054]	[1.000039, 1.000047]
Nb. years in same community	0.99086***	0.98881***	0.99298***	0.99067***	0.98922***	0.99272***
	[0.99002, 0.99170]	[0.98777, 0.98986]	[0.99213, 0.99383]	[0.98984, 0.99149]	[0.98818, 0.99026]	[0.99188, 0.99356]
Share years in same community	1.33***	1.35***	1.23***	1.33***	1.33***	1.24***
	[1.29, 1.38]	[1.31, 1.39]	[1.20, 1.27]	[1.28, 1.37]	[1.29, 1.37]	[1.20, 1.27]
Ln(distance)	0.68***	0.67***	0.68***	0.69***	0.69***	0.69***
	[0.67, 0.69]	[0.66, 0.68]	[0.67, 0.68]	[0.68, 0.70]	[0.68, 0.70]	[0.68, 0.70]
Same empire	1.70***	1.85***	1.84***	1.76***	1.90***	1.89***
	[1.64, 1.77]	[1.79, 1.92]	[1.78, 1.91]	[1.69, 1.83]	[1.84, 1.96]	[1.82, 1.95]
Contiguity	1.44***	1.63***	1.64***	1.35***	1.54***	1.55***
	[1.26, 1.66]	[1.45, 1.82]	[1.47, 1.84]	[1.17, 1.56]	[1.36, 1.73]	[1.38, 1.75]
Conflict (MID)	0.52***	0.58***	0.59***	0.57***	0.65***	0.66***
	[0.42, 0.64]	[0.49, 0.69]	[0.49, 0.70]	[0.47, 0.70]	[0.55, 0.77]	[0.56, 0.78]
Alliance (ATOP)	3.35***	3.91***	3.93***	2.65***	3.08***	3.09***
	[3.01, 3.73]	[3.59, 4.27]	[3.60, 4.29]	[2.41, 2.91]	[2.86, 3.32]	[2.87, 3.34]
Trade agreement	3.27***	3.87***	3.90***	3.13***	3.75***	3.78***
	[2.70, 3.95]	[3.34, 4.47]	[3.37, 4.52]	[2.60, 3.78]	[3.24, 4.33]	[3.27, 4.37]
Observations	4,147,820	6,190,724	6,190,724	3,992,075	6,030,436	6,030,436
Pairs	161,646	240,905	240,905	160,402	240,117	240,117
Log-likelihood	−360752.8	−412705.3	−412667.2	−371911.1	−427428.3	−427334.1

*** p<.01, ** p<.05, * p<.1 -- 95 % confidence intervals in brackets.

Duration dependence is mildly negative over the first three decades (hazard about 3% lower than in the first year) and turns positive for longer durations, although the latter is estimated on fewer observations.

	Start trade (excluding very small flows and failed splits) Time since appearance with zero trade				
	LouvainMaxtrade	IntraMax	Tibi	LouvainTibirescale	AN
Duration (years)	0.9976***	0.9980***	0.9970***	0.9969***	0.9957***
	[0.9970, 0.9982]	[0.9974, 0.9987]	[0.9963, 0.9976]	[0.9963, 0.9976]	[0.9951, 0.9963]
Duration²	1.000042***	1.000040***	1.000044***	1.000043***	1.000046***
	[1.000037, 1.000047]	[1.000035, 1.000045]	[1.000039, 1.000049]	[1.000038, 1.000048]	[1.000041, 1.000051]
Nb. years in same community	0.99086***	0.99048***	0.99120***	0.99147***	0.99923**
	[0.99002, 0.99170]	[0.98957, 0.99139]	[0.99034, 0.99206]	[0.99047, 0.99246]	[0.99856, 0.99989]
Share years in same community	1.33***	1.49***	1.15***	1.30***	0.81***
	[1.29, 1.38]	[1.43, 1.54]	[1.10, 1.19]	[1.24, 1.35]	[0.78, 0.84]
Ln(distance)	0.68***	0.68***	0.67***	0.67***	0.63***
	[0.67, 0.69]	[0.67, 0.69]	[0.66, 0.68]	[0.66, 0.69]	[0.61, 0.64]
Same empire	1.70***	1.69***	1.70***	1.71***	1.73***
	[1.64, 1.77]	[1.63, 1.76]	[1.64, 1.77]	[1.64, 1.78]	[1.66, 1.81]
Contiguity	1.44***	1.45***	1.44***	1.43***	1.46***
	[1.26, 1.66]	[1.27, 1.66]	[1.25, 1.66]	[1.24, 1.64]	[1.26, 1.70]
Conflict (MID)	0.52***	0.52***	0.52***	0.52***	0.52***
	[0.42, 0.64]	[0.43, 0.65]	[0.42, 0.64]	[0.42, 0.64]	[0.42, 0.64]
Alliance (ATOP)	3.35***	3.29***	3.35***	3.36***	3.32***
	[3.01, 3.73]	[2.96, 3.66]	[3.01, 3.73]	[3.02, 3.74]	[2.98, 3.70]
Trade agreement	3.27***	3.22***	3.24***	3.25***	3.05***
	[2.70, 3.95]	[2.66, 3.90]	[2.68, 3.92]	[2.69, 3.93]	[2.50, 3.72]
Observations	4,147,820	4,147,820	4,147,820	4,147,820	4,147,820
Pairs	161,646	161,646	161,646	161,646	161,646
Log-likelihood	−360752.8	−360692.3	−360788.2	−360871.9	−360797.4

*** p<.01, ** p<.05, * p<.1 95 % confidence intervals in brackets.
dur3 = dur_n_fs for flag_trade_entry_n_fs

Membership in the same community raises the hazard of starting trade, by up to 33% for a pair that has always been together, but this advantage declines by about 0.9% for each additional year and disappears after about three decades

CONCLUSION

Outline

1. Data
2. Identifying trade communities
3. Finding the determinants of trade communities
4. Finding the effect of trade communities

Tentative findings

1. Chronology of the importance of trade communities
2. They are not pure artefacts

Decline of the importance of trade communities till the 1890s, stability till WWI, then increase till WWII. Decline since WWII, with not much happening since.

No sign of them being "building blocs"

They play a causal role in the formation of new trade links,

Pat on the back
Much yet to be done

Interdisciplinary work between history, economics and network science

More primary sources on trade statistics -- paradoxically, mainly for post-45

Improve the classification algorithm:

a) Optimise signed Louvain

b) Take ambiguity in Louvain into account

Explore non-linearity

- RICardo: [https://ricardo.medialab.sciences-po.fr]
- GeoPolHist: https://medialab.github.io/GeoPolHist